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B.Tech. (EEE) 4th Semester F-Scheme

Examination, May-2019

TRANSMISSION & DISTRIBUTION

Paper-EE-212-F

Time allowed : 3 hours] *[Maximum marks : 100*

Note : *(i)* ***Question No. 1 (Section-A) is compulsory.***

*(ii) Attempt **four** more questions from remaining **four** Sections (B, C, D and E) by selecting at least **one** question from each section.*

(iii) Use of nonprogrammable scientific calculator is allowed.

Section-A

1. (a) What do you mean by auxiliary supply in power system ? 5

(b) Discuss proximity effect in brief. 5

(c) Explain the use of dampers. 5

(d) Discuss the ratings of cables in brief. 5

Section-B

2. Discuss the types if transmission lines depending on length. Draw and explain the equivalent circuit and phasor diagram of short transmission line. 20

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3. Draw single line diagram of power system. Explain Indoor and outdoor substation in detail. 20

Section-C

4. Discuss and explain different types of Distribution system in detail. 20

5. Derive an expression for the capacitance per km of a single phase line taking into account the effect of ground. https://www.mdustudy.com 20

Section-D

6. What do you mean by Grading of Cables ? Discuss different types of Grading in detail. 20

7. An Overhead line at a river crossing is supported from two towers of height 30 m and 90 m above water level with a span of 300 m. The weight of conductor is 1 kg/m and working tension is 2000 kg. Determine clearance between conductor and water level midway between the towers 20

Section-E

8. What is the use of Insulators in power system ? Explain different types of insulators used for overhead lines in power system. 20

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9. Find the critical distruptive voltage and critical voltage for local and general corona on a 3-phase overhead transmission line, consisting of three stranded copper conductors spaced 2.5 m apart at the corners of an equivalent triangle. Air temperature and pressure are 21° C and 73.6 cm Hg respectively. The conductor dia, irregularity factor and surface factors are 10.4 mm, 0.85, 0.7 and 0.8 respectively. 20

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